PDQ: the week in numbers

4 October 2026. Nada.

	before	after
wrong answers (125 clips)	7	**1**
correct answers	52	48
mirrored clips identified	0 of 8	**8 of 8**
letterboxed clips identified	1 of 8	**6 of 8**

1. What was working before

- The pod's engine, as an exact laptop copy. 52 of 125 known clips right, 7 wrong.
- Three parts can answer, in order: pre-scan, probe (P0), then the rules.

decider	evidence it needs	checks offsets?	share of answers
pre-scan	3 frames, best under 20	no	most
probe P0	1 frame under 30	no	some
rules P1-P9	1 to 30 frames by rule	only the mid rules	the rest

2. How it was working

- The quick parts decide most clips on counting alone. The careful checks sit in rules that rarely decide.
- A true match gives the same offset (film time minus clip time) for every frame. Nothing that decides most of the traffic checks this.
- Every answer now records the rule that made it. The seven wrong answers:

rule	wrong answers	distance
pre-scan	4 (Krrish 3, Northman, Civil War, In Between)	6 - 18
P0 probe	1 (Smile)	26.8
P9E	1 (Grave Encounters 2)	34.0
P8	1 (Demeter)	41.4

```
PS C:\Users\nadaa\pdq_for_nada> python attribute_rules.py --engine pdq_engine_ORIGINAL.py

================================================================================
  WHICH RULE PRODUCED THE ANSWERS   (engine: pdq_engine_ORIGINAL.py)
================================================================================

  WRONG answers reproduced (these rules are the problem): 7
     SCAN       4
     P0         1
     P9E        1
     P8         1
       SCAN       Krrish 3                          avg=6.0 frames=27
       P0         Smile                             avg=26.8 frames=5
       SCAN       Captain America: Civil War        avg=12.0 frames=4
       P9E        Grave Encounters 2                avg=34.0 frames=2
       SCAN       The In Between                    avg=18.0 frames=4
       SCAN       The Northman                      avg=6.0 frames=6
       P8         The Last Voyage of the Demeter    avg=41.4 frames=31

  CORRECT answers (these rules must keep working): 52
     SCAN       46
     P0         3
     P3.5       2
     P1         1
```

• Four of seven come from the pre-scan. Rule changes can never reach them.

3. What I built

• **Attribution testing**: every answer tagged with the rule that produced it.

• **A rebuilt probe**: whole-clip view + the careful checks, as one decider.

• **An edits test**: 10 reposter edits x 8 clips that work unedited.

• **Two fixes**: mirrored views, and cutting black bars before fingerprinting.

4. How the new probe works

• Samples 40 frames across the whole clip; keeps a scorecard per film: frames, best distance, and every hit's (clip time, film time) pair.

• Answers only when the evidence agrees like a real scene does:

bar	needs	copied from
PV-A	2 near-identical frames	P2.5
PV-B	3 frames, best under 20, 2 agreeing on offset	pre-scan + P4
PV-C	5 medium frames agreeing, from 2 clip moments	P6a

• The one-frame P0 is retired. The rules stay behind it as backup.

Proof, on real clips. Both pass the OLD pre-scan bar; only the agreement question separates them:

```
PS C:\Users\nadaa\pdq_for_nada> python probe_walkthrough.py <the Northman clip> Northman
clip: vm_tiktok_com_ZS4MtfV2y_This_post_is_shared_via_TikTok_Lite_Download_TikTok_Lite.mp4
film examined: 290:The Northman (tmdb=639933)

THE PROBE'S EVIDENCE for this film (6 sampled frames hit it):
  clip time   film time   distance      offset
    0.74min    49.62min          6    48.88min
    0.36min    40.31min         16    39.95min
    0.41min    43.39min         20    42.98min
    0.52min    49.13min         26    48.61min
    0.25min    36.96min         30    36.71min
    0.30min    37.02min         44    36.72min

THE THREE BARS:

  PV-A  two near-identical frames?
        best=6 (needs <12)   frames under 16: 1 (needs 2)
        -> FAIL

  PV-B  decent frames that AGREE on film position?
        best=6 (needs <20)   frames under 28: 4 (needs 3)
        hits agreeing on one offset: 1 (needs 2)
        -> FAIL

  PV-C  mediocre frames, proven twice over?
        best=6 (needs <26)   frames under 33: 5 (needs 5)
        agreeing: 1 (needs 3)   clip moments: 5 (needs 2)
        -> FAIL

VERDICT of the probe for this film: STAYS QUIET - the rules (or nothing) take over
```

```
PS C:\Users\nadaa\pdq_for_nada> python probe_walkthrough.py <the Brick clip> Brick
clip: vm_tiktok_com_ZN8rNvaKw_.mp4
film examined: 202981:Brick (tmdb=9270)

THE PROBE'S EVIDENCE for this film (13 sampled frames hit it):
  clip time   film time   distance      offset
    0.31min    97.10min          8    96.80min
    0.13min    42.59min         16    42.46min
    0.34min    88.16min         16    87.83min
    0.36min    67.31min         18    66.95min
    0.08min    19.14min         20    19.06min
    0.12min    19.17min         26    19.06min
    0.28min    17.12min         36    16.84min
    0.30min    74.77min         36    74.47min
    0.01min    29.46min         40    29.45min
    0.03min    19.09min         42    19.06min
    0.29min    17.15min         42    16.87min
    0.38min    45.03min         42    44.65min
    0.04min    19.09min         44    19.05min

THE THREE BARS:

  PV-A  two near-identical frames?
        best=8 (needs <12)   frames under 16: 1 (needs 2)
        -> FAIL

  PV-B  decent frames that AGREE on film position?
        best=8 (needs <20)   frames under 28: 6 (needs 3)
        hits agreeing on one offset: 2 (needs 2)
        -> PASS

  PV-C  mediocre frames, proven twice over?
        best=8 (needs <26)   frames under 33: 6 (needs 5)
        agreeing: 2 (needs 3)   clip moments: 2 (needs 2)
        -> FAIL

VERDICT of the probe for this film: ANSWER: 202981:Brick (tmdb=9270)
```

And the honest third case: a true scene with thin evidence. Perfect agreement (every offset 32.2) but too few frames, so the probe declines - and the fallback rules then name Soul via P1. Declined is not lost:

```
PS C:\Users\nadaa\pdq_for_nada> python probe_walkthrough.py <the Soul clip> Soul
clip: www_instagram_com_reel_DZ_L9fDykBP_.mp4
film examined: 265:Soul (tmdb=508442)

THE PROBE'S EVIDENCE for this film (5 sampled frames hit it):
  clip time   film time   distance      offset
    0.01min    32.24min          8    32.24min
    0.07min    32.29min         16    32.23min
    0.04min    32.24min         32    32.20min
    0.06min    32.27min         34    32.22min
    0.09min    32.31min         42    32.22min

THE THREE BARS:

  PV-A  two near-identical frames?
        best=8 (needs <12)   frames under 16: 1 (needs 2)
        -> FAIL

  PV-B  decent frames that AGREE on film position?
        best=8 (needs <20)   frames under 28: 2 (needs 3)
        hits agreeing on one offset: 2 (needs 2)
        -> FAIL

  PV-C  mediocre frames, proven twice over?
        best=8 (needs <26)   frames under 33: 3 (needs 5)
        agreeing: 3 (needs 3)   clip moments: 1 (needs 2)
        -> FAIL

VERDICT of the probe for this film: STAYS QUIET - the rules (or nothing) take over
```

5. Results

Real clips, all 125:

engine	wrong	correct
pod's rules	7	52
new probe alone	1	41
new probe + rules backup	**1**	**48**

```
PS C:\Users\nadaa\pdq_for_nada> python attribute_rules.py --engine pdq_engine_PROBEV2.py

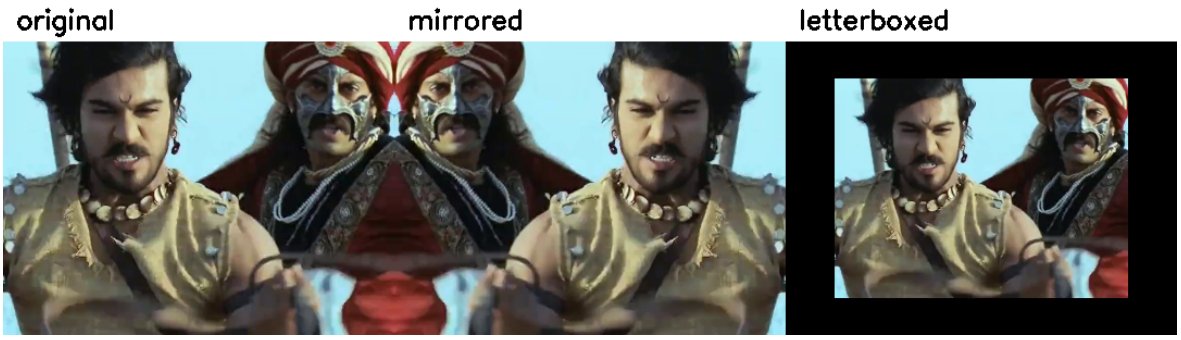
================================================================================
  WHICH RULE PRODUCED THE ANSWERS   (engine: pdq_engine_PROBEV2.py)
================================================================================

  WRONG answers reproduced (these rules are the problem): 1
     PV-A       1
       PV-A       Krrish 3                            avg=6.0 frames=16

  CORRECT answers (these rules must keep working): 48
     PV-A       22
     PV-B       17
     P1         4
     P3.5       3
     PV-C       2
```

- The 1 left: Krrish 3, distance 6, identical pictures. Needs audio / SSCD.
- The 4 lost: fan edits that jump scenes. Recovering them readmits 2 wrongs. Kept strict.

Edited clips:



edit	before	after
mirror	0 of 8	**8 of 8** (distance 2)
letterbox	1 of 8	**6 of 8**
filter / b&w / recompress / speeds	6-8 of 8	unchanged

- No edit ever produced a wrong film, on any version. Edits silence, never fool.
- Safety gate: all 125 real clips re-run with the fixes. Zero answers changed.
- Cost: probe 4x slower with extra views. Pod shape: normal views first.

The whole story in three tables

Before: what was working, and how

part	what it is	how it decides	its results (125 clips)
pre-scan	a quick skim before anything else: 40 frames spread across the whole clip	counts hits: 3 frames under distance 28, best under 20. Never asks whether hits agree on film position or come from several clip moments	most answers: 46 correct, but also 4 of the 7 wrong (Krrish 3, Northman, Civil War, In Between, at distances 6-18)
probe P0	one quick look at the first 4 seconds, one crop shape only	a single frame under distance 30 is enough to answer	3 correct, 1 wrong (Smile, answered off one frame)
rules P1-P9	scorecards per film, watched as the clip is read; first rule to fire wins	the careful mid rules check agreement and clip spread; the weak ones (P8, P9E) mostly count mediocre frames	3 correct, 2 wrong (Demeter via P8, Grave Encounters 2 via P9E)
all together	quick parts answer first, so the careful checks are rarely reached	counting decides most of the traffic	**52 correct, 7 wrong**

After: what I built, and how it works

built	how it works	measured result
attribution testing	every clip run records which rule produced the answer	each of the 7 wrong answers pinned to its rule; every later change judged by re-running all 125 clips
the probe V2	40 frames across the whole clip; each fingerprinted in every view (full, caption-trimmed, all crops, mirrored, bars removed); a scorecard per film with every hit's film position	on its own: 1 wrong, 41 correct
bar PV-A (from P2.5)	two near-identical frames, under ~distance 12-16	lets in 22 of the 41
bar PV-B (pre-scan + P4)	three decent frames, best under 20, AND two agreeing on film position	lets in 17; refuses Northman, Civil War, In Between
bar PV-C (from P6a)	five medium frames, three agreeing, from two clip moments	lets in 2 (heavily filtered clips)
P0 retired, P9E floor	no more one-frame answers; two frames only count for P9E when the average is 25 or better	removes Smile and Grave Encounters 2
the edits harness + 2 fixes	10 reposter edits x 8 clips; probe now fingerprints mirrored views and cuts black bars first	mirror 0/8 to 8/8 (distance 2); letterbox 1/8 to 6/8; zero changes on real clips
drop-in mode	the probe decides what it trusts; everything it declines falls through to the rules, which read the clip densely	**1 wrong, 48 correct**

Before against after

	before	after
wrong answers	7	**1** (Krrish 3: identical pictures, needs audio/SSCD)
correct answers	52	48 (the 4 lost are fan edits - the policy question)
agreement checked where most decisions happen	no	yes
minimum evidence for an answer	1 frame	2 near-identical frames
mirrored clips identified	0 of 8	**8 of 8**
letterboxed clips identified	1 of 8	**6 of 8**
wrong films caused by edits	0	0
every answer names its rule	no	yes
probe cost per clip	~30 s	~2 min (normal-views-first restores it)

The asks

1. Policy call: the 4 fan-edit clips. I recommend staying strict.
2. Someone watches the Misfits clip (both engines say F1, distance 8).
3. Direction: take the probe towards the pod? Next step: normal-views-first at pod scale.
4. Still open: keep clip_query_hashes rows for voted clips. Clips are still rotting.